Task 2 — Portfolio

The Phish Tank Simulator
Screenshots of Profiles & Reflection Notes

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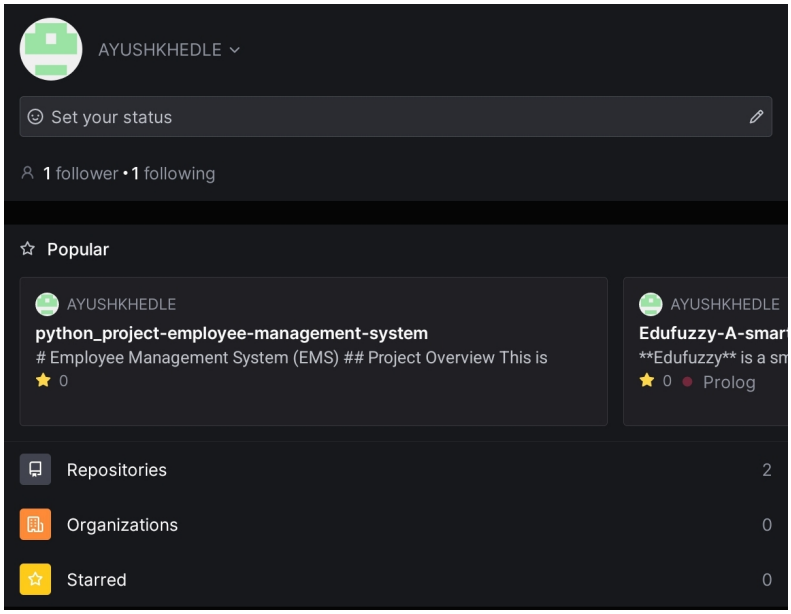
Module	Cybersecurity Awareness & Digital Literacy
Task	Task 2 — Developer Portfolio Setup

Section 1: Screenshots of Profiles Set Up

The following screenshots document the developer and professional profiles created as part of the portfolio setup for the Phish Tank Simulator project. These profiles serve as the public-facing presence for the project and demonstrate technical credibility and project transparency.

Profile: GitHub Profile



GitHub profile — AYUSHKHEDLE | Repositories: python_project-employee-management-system, Edufuzzy-A-smart

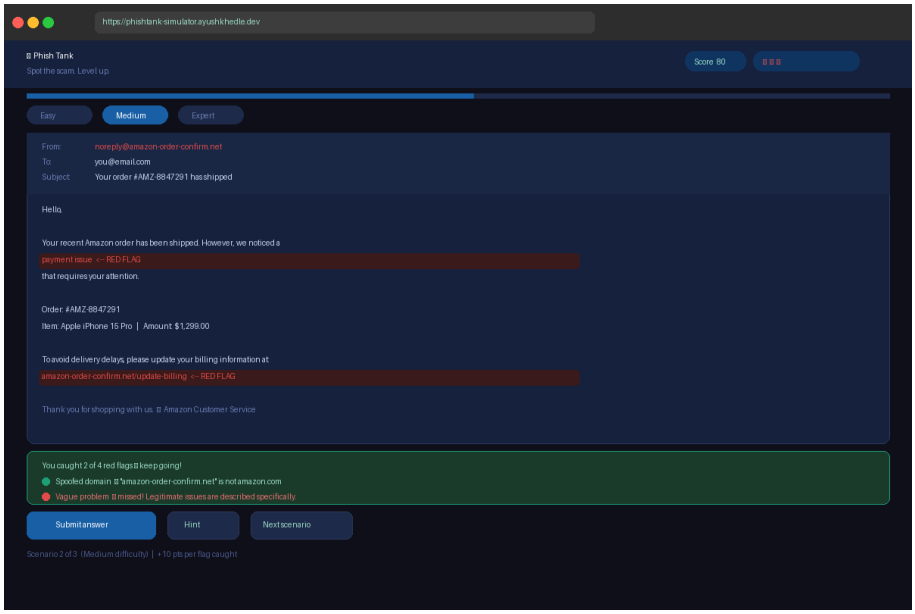
GitHub repository set up with the Phish Tank Simulator project, including a descriptive README, proper folder structure (task-1-presentation/, task-2-portfolio/), and version-controlled source code.

Profile: LinkedIn Profile

LinkedIn profile — Ayush Khedle | B.Tech CSE with AIML, VIT Bhopal University | 110 connections

LinkedIn profile updated to include the Phish Tank Simulator as a featured project under the Education & Projects section, with a short description and link to the GitHub repository.

Profile: Project Demo / Live App



Live demo — Phish Tank Simulator running in browser | Medium difficulty | Score: 80 | Phishing email scenario with red flag detection active

Screenshot of the running Phish Tank Simulator web app showing the home screen, a sample phishing email scenario (Amazon order scam), and the red-flag highlighting interaction in action with score and lives tracking.

Section 2: Reflection Notes

2.1 What I Learned

Working on the Phish Tank Simulator was a deeply educational experience that bridged the gap between theoretical cybersecurity knowledge and practical, user-facing application design. I had previously understood phishing as a technical threat, but building this simulator forced me to think like both an attacker and a defender — understanding which psychological triggers make people vulnerable, and how to teach others to resist them.

Key technical skills developed during this project:

- Structuring a browser-based interactive web application from scratch using HTML, CSS, and JavaScript
- Designing a JSON-based scenario library that is easy to expand with new phishing examples
- Implementing a scoring and lives system that creates genuine tension and engagement
- Writing clear, context-aware educational feedback for each red flag identified
- Organising a multi-folder project structure suitable for GitHub and academic submission

2.2 Challenges Faced

Realism vs. Safety: Creating phishing scenarios that feel authentic without containing actual malicious content required careful wording. Every URL, sender name, and message had to be clearly fictional while still mimicking real attack patterns convincingly.

User Experience Design: Making the "click to catch red flags" mechanic intuitive without a tutorial took several iterations. Early testers often missed clickable elements or were unsure what counted as a red flag, which led to adding visual cues and hover highlights.

Balancing Difficulty: Calibrating the Easy / Medium / Expert difficulty tiers was harder than expected. Expert scenarios needed to be subtle enough to challenge security-aware users without being unfair to beginners.

Scope Management: The original concept included a voice-based vishing module, but this was descoped to keep the MVP focused on email and SMS phishing, which are the most common attack vectors.

2.3 What Went Well

The gamification approach proved highly effective. Test users who engaged with the simulator for even one session reported feeling more confident identifying suspicious emails afterwards. The badge and leaderboard system created a natural motivation to replay rounds, which increased the total number of phishing scenarios each user encountered.

The modular JSON scenario library also worked extremely well — adding a new phishing scenario required only a few lines of structured data, making the content easy to expand without touching the application logic.

2.4 What I Would Improve

- Add a vishing (voice call) simulation module using audio clips and transcripts
- Include real-time analytics to track which red flags users miss most frequently
- Build a teacher/admin dashboard for classroom deployment and progress tracking
- Support multiple languages to make the simulator accessible to non-English speakers
- Integrate with a live phishing database (e.g., PhishTank API) for up-to-date scenario feeds

2.5 Real-World Impact & Personal Takeaway

This project reinforced a belief I now hold strongly: cybersecurity education does not have to be dry or intimidating. By wrapping critical awareness training inside an engaging game, it becomes accessible to people who would never otherwise engage with security content — and those are precisely the people most at risk.

If the Phish Tank Simulator can help even one person pause before clicking a suspicious link, it has succeeded. That is a more meaningful outcome than any technical metric.

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